

Calculus 3

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May 8, 2024

Introduction

Course description:

- The course is designed to equip the students with the concepts, techniques, and applications of calculus of several variables, including partial differentiation, multiple integration, and vector calculus.
- The course provides mathematical background for courses in Computer Science. The course emphasizes all three aspects: conceptual understanding, computational and problem solving ability, and exposure to applications.

References:

- *Giáo trình phép tính vi tích phân*,
<https://sites.google.com/view/math-hcmus-edu-vn-giaitich>
- *Calculus, Concepts and Contexts*, James Stewart, Thompson Brooks/Cole, 7th Edition, 2012.

Softwares:

- Mathematical computation softwares: Maxima, or Maple, or Matlab, or Wolfram Alpha, or GeoGebra...
- Software for typesetting mathematics: LaTeX, or LyX.

1 Differentiation

- Spaces
- Functions of several variables
- Partial derivatives
- Tangent Planes
- Linear Approximations
- The Chain Rule
- Directional derivative
- Extremum of functions of several variables
- Lagrange multipliers

1 Differentiation

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Differentiation - Spaces

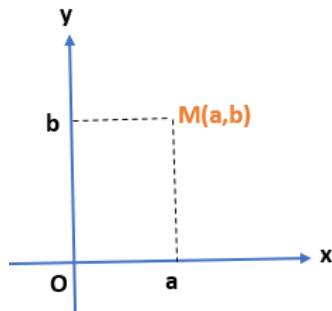


Figure: A point in a plane.

Any point in the plane can be represented as an ordered pair (a, b) of real numbers, where

- a is the x -coordinate.
- b is the y -coordinate

Differentiation - Spaces

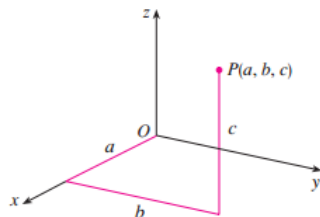


Figure: A point in space.

Any **point in space** can be represented as an ordered pair (a, b, c) of real numbers, where

- a is the x -coordinate $\Rightarrow$ the distance from the yz -plane to P .
- b is the y -coordinate $\Rightarrow$ the distance from the xz -plane to P .
- c is the z -coordinate $\Rightarrow$ the distance from the xy -plane to P

Differentiation - Spaces

Remark:

- In two-dimensional analytic geometry, the graph of an equation involving x and y is a **curve** in $\mathbb{R}^2$.
- In three-dimensional analytic geometry, an equation in x , y , and z represents a **surface** in $\mathbb{R}^3$.

Example:

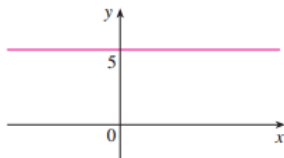


Figure: $y = 5$, a line in $\mathbb{R}^2$.

The equation $y = 5$ represents the set $\{(x, y) | y = 5\}$, which is the set of all points in $\mathbb{R}^2$ whose y -coordinate is 5 (x can each be any value).

Differentiation - Spaces

Example:

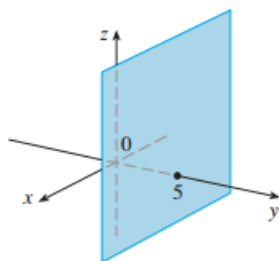


Figure: $y = 5$, a line in $\mathbb{R}^2$.

The equation $y = 5$ represents the set of all points in $\mathbb{R}^3$ whose y -coordinate is 5. This is the vertical plane that is parallel to the xz -plane and five units to the right of it.

Differentiation - Spaces

Example: What does the equation $x^2 + y^2 = 1$ represent as a surface in $\mathbb{R}^3$?

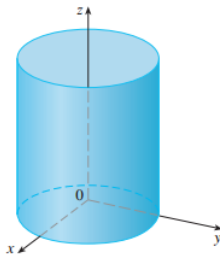


Figure: The cylinder $x^2 + y^2 = 1$.

Given that $x^2 + y^2 = 1$, with no restrictions on z , we can see that the point (x, y, z) could lie on a circle in any horizontal plane $z = k$. So the surface $x^2 + y^2 = 1$ in $\mathbb{R}^3$ consists of all possible horizontal circles $x^2 + y^2 = 1, z = k$, and is therefore the circular cylinder with radius 1 whose axis is the z -axis.

Differentiation - Spaces

Example: Describe and sketch the surface in $\mathbb{R}^3$ represented by the equation $y = x$.

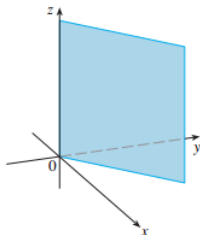


Figure: The plane $y = x$.

The equation represents the set of all points in $\mathbb{R}^3$ whose x - and y -coordinates are equal, that is, $\{(x, x, z) | x \in \mathbb{R}, z \in \mathbb{R}\}$. This is a vertical plane that intersects the xy -plane in the line $y = x, z = 0$. The portion of this plane that lies in the first octant.

Differentiation - Combining vectors

Definition of vector addition

If $\mathbf{u}$ and $\mathbf{v}$ are vectors positioned so the initial point of $\mathbf{v}$ is at the terminal point of $\mathbf{u}$, then the sum $\mathbf{u} + \mathbf{v}$ is the vector from the initial point of $\mathbf{u}$ to the terminal point of $\mathbf{v}$.

The definition of vector addition is illustrated in the following figure

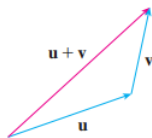
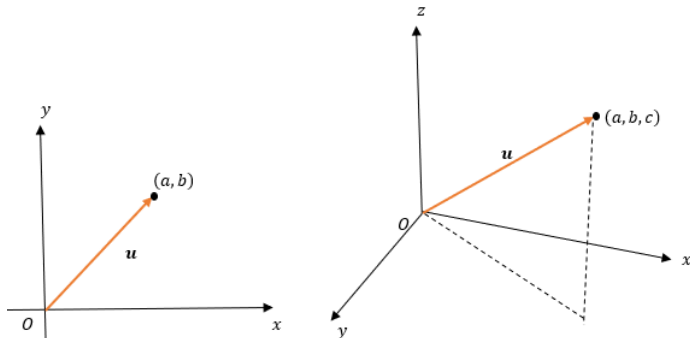


Figure: The Triangle Law

Remark: The definition of vector addition is sometimes called the **Triangle Law**.

Differentiation - Component

- In **two-dimensional** (or **three-dimensional**), we place the initial point of a vector $\mathbf{u}$ at the origin of a rectangular coordinate system, then the terminal point of $\mathbf{u}$ has coordinates of the form (a, b) (or (a, b, c)).



Differentiation - Component

- Given the points $A(a_1, b_1)$ and $B(a_2, b_2)$ in $\mathbb{R}^2$, the vector $\overrightarrow{AB}$ is

$$\overrightarrow{AB} = \langle a_2 - a_1, b_2 - b_1 \rangle. \quad (1)$$

- Given the points $A(a_1, b_1, c_1)$ and $B(a_2, b_2, c_2)$ in $\mathbb{R}^3$, the vector $\overrightarrow{AB}$ is

$$\overrightarrow{AB} = \langle a_2 - a_1, b_2 - b_1, c_2 - c_1 \rangle. \quad (2)$$

Example: Find the vector represented by the directed line segment with initial point $A(2, -3, 4)$ and terminal point $B(-2, 1, 1)$.

Differentiation - The magnitude or length of the vector

The magnitude or length of the vector

- The **magnitude or length** of the vector $\mathbf{u} = \langle a, b \rangle$ in two-dimensional is

$$|\mathbf{u}| = \sqrt{a^2 + b^2}. \quad (3)$$

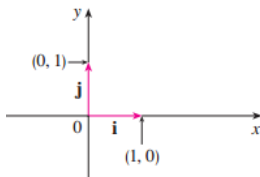
- The **magnitude or length** of the vector $\mathbf{u} = \langle a, b, c \rangle$ in three-dimensional is

$$|\mathbf{u}| = \sqrt{a^2 + b^2 + c^2}. \quad (4)$$

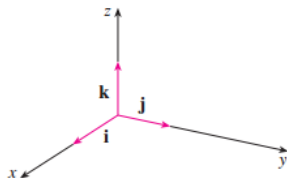
Example: Let the vector $\mathbf{u} = \langle 4, 0, 3 \rangle$. Find $|\mathbf{u}|$.

Differentiation - Standard basis vectors

- In two-dimensional, these vectors $\mathbf{i} = \langle 1, 0 \rangle$ and $\mathbf{j} = \langle 0, 1 \rangle$ are called the **standard basis vectors**. They have length 1 and point in the directions of the positive x - and y -axes.



- In three-dimensional, these vectors $\mathbf{i} = \langle 1, 0, 0 \rangle$, $\mathbf{j} = \langle 0, 1, 0 \rangle$, and $\mathbf{k} = \langle 0, 0, 1 \rangle$ are called the **standard basis vectors**. They have length 1 and point in the directions of the positive x -, y -, and z -axes.



Differentiation - Standard basis vectors

- In two-dimensional, we can write

$$\mathbf{u} = \langle a, b \rangle = a\mathbf{i} + b\mathbf{j}.$$

- In three-dimensional, we can write

$$\mathbf{u} = \langle a, b, c \rangle = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}.$$

Unit vector

A **unit vector** is a vector whose length is 1.

Example: $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are all unit vectors.

Remark: In general, if $\mathbf{v} \neq 0$, then the unit vector that has the same direction as $\mathbf{v}$ is

$$\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|}.$$

Differentiation - The dot product

The dot product

If $\mathbf{u} = \langle a_1, b_1, c_1 \rangle$ and $\mathbf{v} = \langle a_2, b_2, c_2 \rangle$, then the **dot product** of $\mathbf{u}$ and $\mathbf{v}$ is the number $\mathbf{u} \cdot \mathbf{v}$ given by

$$\mathbf{u} \cdot \mathbf{v} = a_1 a_2 + b_1 b_2 + c_1 c_2.$$

Example: The dot product of $\mathbf{u} = \langle 2, 4 \rangle$ and $\mathbf{v} = \langle 3, -1 \rangle$ is

$$\mathbf{u} \cdot \mathbf{v} = 2 \times 3 + 4 \times (-1) = 2.$$

Differentiation - The cross product

Let two nonzero vectors $\mathbf{u} = \langle a_1, b_1, c_1 \rangle$ and $\mathbf{v} = \langle a_2, b_2, c_2 \rangle$.

Problem: Find a nonzero vector $\mathbf{w}$ that is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$?

Solution: Assume that $\mathbf{w} = \langle a_3, b_3, c_3 \rangle$. Since $\mathbf{w}$ is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$, we have

$$\begin{cases} \mathbf{u} \cdot \mathbf{w} = 0 \\ \mathbf{v} \cdot \mathbf{w} = 0 \end{cases} \Leftrightarrow \begin{cases} a_1 a_3 + b_1 b_3 + c_1 c_3 = 0 \\ a_2 a_3 + b_2 b_3 + c_2 c_3 = 0 \end{cases} \quad (5)$$

To eliminate c_3 , we multiply the first equation in (5) by c_2 and the second equation in (5) by c_1 and subtract:

$$(a_1 c_2 - a_2 c_1) a_3 + (b_1 c_2 - b_2 c_1) b_3 = 0. \quad (6)$$

So, we choose

$$a_3 = b_1 c_2 - b_2 c_1, \quad b_3 = -a_1 c_2 + a_2 c_1.$$

Substituting these values into (5), we have

$$c_3 = a_1 b_2 - a_2 b_1.$$

Differentiation - The cross product

This means that a vector perpendicular to both $\mathbf{u}$ and $\mathbf{v}$ is

$$\mathbf{w} = \langle a_3, b_3, c_3 \rangle = \langle b_1 c_2 - b_2 c_1, -a_1 c_2 + a_2 c_1, a_1 b_2 - a_2 b_1 \rangle.$$

The resulting vector $\mathbf{w}$ is called the **cross product** of $\mathbf{u}$ and $\mathbf{v}$ and is denoted by $\mathbf{u} \times \mathbf{v}$.

Cross product

If $\mathbf{u} = \langle a_1, b_1, c_1 \rangle$ and $\mathbf{v} = \langle a_2, b_2, c_2 \rangle$, then the **cross product** of $\mathbf{u}$ and $\mathbf{v}$ is **the vector**

$$\mathbf{u} \times \mathbf{v} = \langle b_1 c_2 - b_2 c_1, -a_1 c_2 + a_2 c_1, a_1 b_2 - a_2 b_1 \rangle.$$

Remark: $\mathbf{u} \times \mathbf{v}$ is defined only when $\mathbf{u}$ and $\mathbf{v}$ are *three-dimensional* vectors. In order to make the definition of the cross product of $\mathbf{u}$ and $\mathbf{v}$ easier to remember, we use the notation of determinants. We often write

$$\mathbf{u} \times \mathbf{v} = \left| \begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{pmatrix} \right|.$$

Differentiation - The cross product

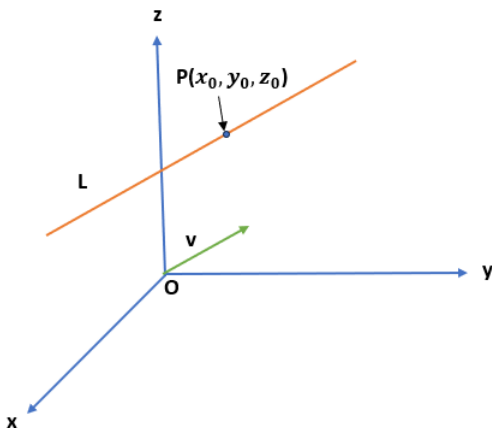
Example: If $\mathbf{u} = \langle 1, 3, 4 \rangle$ and $\mathbf{v} = \langle 2, 7, -5 \rangle$ then

$$\begin{aligned}\mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 3 & 4 \\ 2 & 7 & -5 \end{vmatrix} \\ &= \begin{vmatrix} 3 & 4 \\ 7 & -5 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 4 \\ 2 & -5 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 3 \\ 2 & 7 \end{vmatrix} \mathbf{k} \\ &= (-15 - 28)\mathbf{i} - (-5 - 8)\mathbf{j} + (7 - 6)\mathbf{k} \\ &= -43\mathbf{i} + 13\mathbf{j} + \mathbf{k} \\ &= (-43, 13, 1).\end{aligned}$$

Differentiation - Equations of Lines and Planes

A **line** L in three-dimensional space is determined when we know a point $P_0(x_0, y_0, z_0)$ on L and the direction of L .

In three dimensions the direction of a line is conveniently described by a vector, so we let $\mathbf{v} = \langle a, b, c \rangle$ be a vector parallel to L .

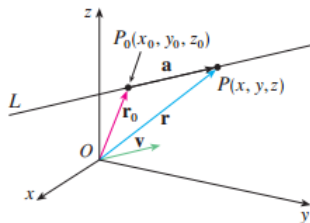


Problem: Find the equation of the line L ???

Differentiation - Equations of Lines and Planes

Let $P(x, y, z)$ be an arbitrary point on L . Then, we define

$$\mathbf{r}_0 = \overrightarrow{OP_0} = \langle x_0, y_0, z_0 \rangle, \quad \mathbf{r} = \overrightarrow{OP} = \langle x, y, z \rangle, \quad \mathbf{a} = \overrightarrow{P_0P}.$$



By using the triangle law for vector addition, we have

$$\mathbf{r} = \mathbf{r}_0 + \mathbf{a}.$$

Since $\mathbf{a}$ and $\mathbf{v}$ are parallel vectors, there exists $t \in \mathbb{R}$ such that

$$\mathbf{a} = t\mathbf{v}.$$

Differentiation - Equations of Lines and Planes

Thus, the vector equation of L is

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v} \Leftrightarrow \langle x, y, z \rangle = \langle x_0, y_0, z_0 \rangle + t \langle a, b, c \rangle .$$

Two vectors are equal if and only if corresponding components are equal. Therefore we have the three scalar equations:

$$\begin{cases} x = x_0 + at \\ y = y_0 + bt \\ z = z_0 + ct \end{cases} \quad (7)$$

where $t \in \mathbb{R}$. These equations are called **parametric equations** of the line L through the point $P_0(x_0, y_0, z_0)$ and parallel to the vector $\mathbf{v} = \langle a, b, c \rangle$.

Remark: Each value of the parameter t gives a point (x, y, z) on L .

Example:

- (a) Find parametric equations for the line that passes through the point $(5, 1, 3)$ and is parallel to the vector $\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$.
- (b) Find two other points on the line.

Differentiation - Equations of Lines and Planes

Another way of describing a line L is to eliminate the parameter t from equation (7). If none of a , b or c is 0, we can solve each of these equations for t :

$$\begin{cases} t = \frac{x - x_0}{a} \\ t = \frac{y - y_0}{b} \\ t = \frac{z - z_0}{c} \end{cases}$$

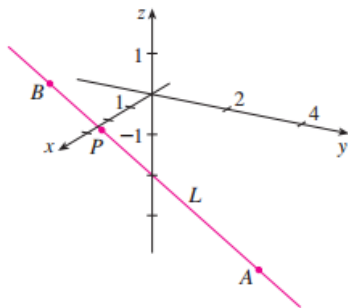
Equating the results, we obtain

Symmetric equations

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}. \quad (8)$$

Differentiation - Equations of Lines and Planes

Example: Find parametric equations and symmetric equations of the line that passes through the points $A(2, 4, -3)$ and $B(3, -1, 1)$.



Differentiation - Equations of Lines and Planes

- A line in space is determined by a point and a direction.
- A plane in space is more difficult to describe. A single vector parallel to a plane is not enough to convey the “direction” of the plane, but a vector perpendicular to the plane does completely specify its direction.

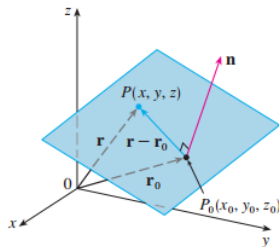
⇒ A plane in space is determined by a point $P_0(x_0, y_0, z_0)$ in the plane and a vector n that is orthogonal to the plane. This orthogonal vector n is called a **normal vector**.

Problem: Find the equation of the plane???

Differentiation - Equations of Lines and Planes

Let $P(x, y, z)$ be an arbitrary point in the plane and $\mathbf{n} = \langle a, b, c \rangle$. Then, we define

$$\mathbf{r}_0 = \overrightarrow{OP_0} = \langle x_0, y_0, z_0 \rangle, \quad \mathbf{r} = \overrightarrow{OP} = \langle x, y, z \rangle,$$
$$\mathbf{r} - \mathbf{r}_0 = \overrightarrow{P_0P} = \langle x - x_0, y - y_0, z - z_0 \rangle.$$



Since $\mathbf{n}$ is the normal vector of the given plane, $\mathbf{n}$ is orthogonal to every vector in the given plane. Thus, $\mathbf{n}$ is orthogonal to $\mathbf{r} - \mathbf{r}_0$ and so we have

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0 \Leftrightarrow a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

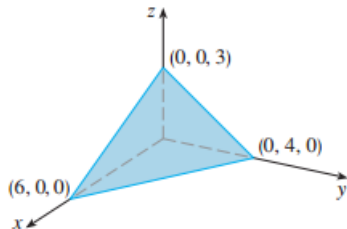
Scalar equation of the plane

Let $P(x, y, z)$ be an arbitrary point in the plane. A **scalar equation of the plane** through point $P_0(x_0, y_0, z_0)$ with normal vector $\mathbf{n} = \langle a, b, c \rangle$ is

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0. \quad (9)$$

Differentiation - Equations of Lines and Planes

Example: Find an equation of the plane through the point $(2, 4, -1)$ with normal vector $\mathbf{n} = \langle 2, 3, 4 \rangle$. Find the intercepts and sketch the plane.



Differentiation - Equations of Lines and Planes

Example: Find an equation of the plane that passes through the points $P(1, 3, 2)$, $Q(3, -1, 6)$ and $R(5, 2, 0)$.

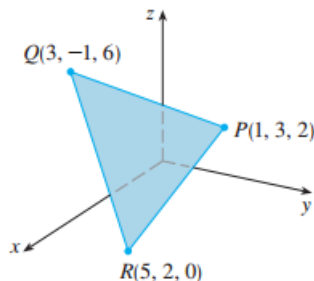


Figure: The portion of the plane that is enclosed by triangle PQR .

Functions of several variables

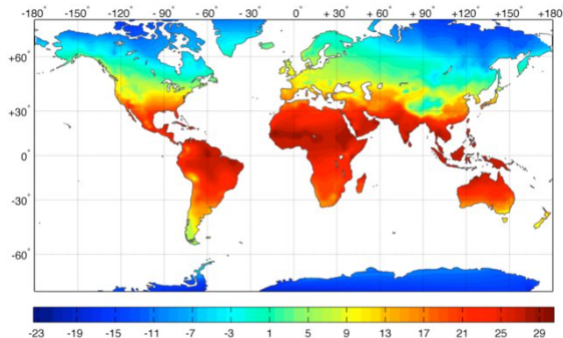


Figure: The temperature on the surface of the earth.

- The temperature T at a point on the surface of the earth at any given time depends on the longitude x and latitude y of the point.
- T is a function of the two variables x and y , or as a function of the pair (x, y) . We indicate this functional dependence by writing $T = f(x, y)$.

Functions of several variables

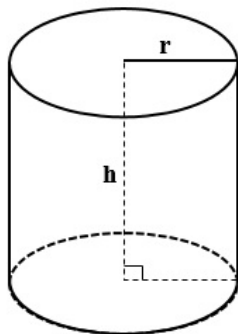


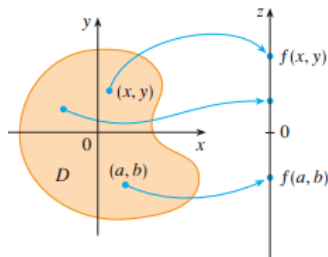
Figure: A circular cylinder.

- The volume V of a circular cylinder depends on its radius r and its height $h \Rightarrow V = \pi r^2 h$.
- V is a function of r and h , and we write $V(r, h) = \pi r^2 h$.

Functions of several variables

Definition:

- A **function of two variables** f is a rule that assigns to each ordered pair of real numbers (x, y) in a set D a unique real number denoted by $f(x, y)$.
- The set D is the **domain** of f and its **range** is the set of values that f takes on, that is, $R = \{f(x, y) : (x, y) \in D\}$.



- We often write $z = f(x, y)$ to make explicit the value taken on by f at the general point (x, y) .
- The variables x and y are **independent variables**.
- z is the **dependent variable**.

Remark

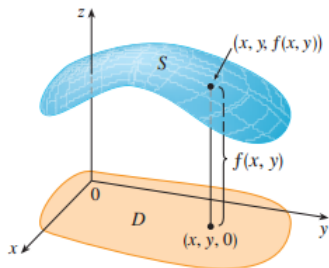
If a function f is given by a formula and no domain is specified, then the domain of f is understood to be the set of all pairs (x, y) for which the given expression is a well-defined real number.

Example: Find the domain and range of $f(x, y) = \sqrt{9 - x^2 - y^2}$.

Functions of several variables

Definition

If f is a function of two variables with domain D , then the **graph** of f is the set of all points (x, y, z) in $\mathbb{R}^3$ such that $z = f(x, y)$ and (x, y) is in D .



- The graph of a function f of two variables is a surface S with equation $z = f(x, y)$.
- The graph S of f as lying directly above or below its domain D in the xy -plane.

Functions of several variables

Example: Sketch the graph of $f(x, y) = \sqrt{9 - x^2 - y^2}$.

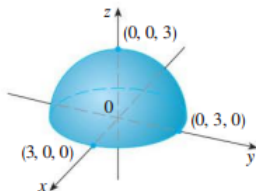


Figure: Graph of $f(x, y) = \sqrt{9 - x^2 - y^2}$.

Definition

The **level curves** of a function f of two variables are the curves with equations $f(x, y) = k$, where k is a constant (in the range of f).

A level curve $f(x, y) = k$ is the set of all points in the domain of f at which f takes on a given value k . In other words, it shows where the graph of f has height k .

Functions of several variables

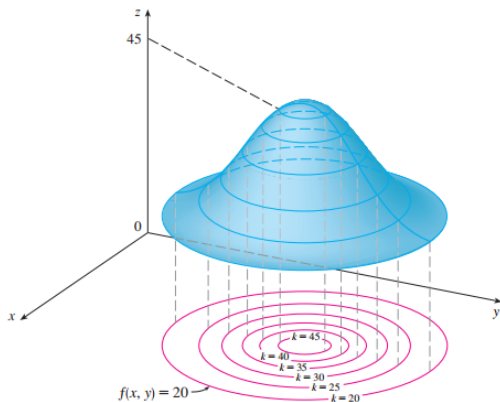


Figure: The relation between level curves and horizontal traces.

- The level curves $f(x, y) = k$ are just the traces of the graph of f in the horizontal plane $z = k$ projected down to the xy -plane. So if you draw the level curves of a function and visualize them being lifted up to the surface at the indicated height, then you can mentally piece together a picture of the graph.
- The surface is steep where the level curves are close together. It is somewhat flatter where they are farther apart.

Functions of several variables

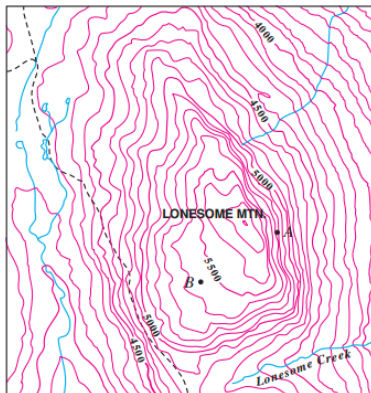
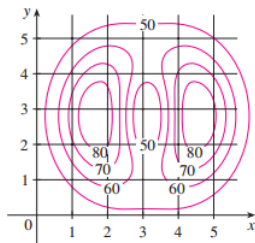


Figure: Level curves occurs in topographic maps of mountainous regions.

- The level curves are curves of constant elevation above sea level.
- If you walk along one of these contour lines, you neither ascend nor descend.

Functions of several variables

Example: A contour map for a function f is shown in the following figure



Use it to estimate the values of $f(1, 3)$ and $f(4, 5)$.

Functions of several variables

Example: Sketch the level curves of the function $f(x, y) = 6 - 3x - 2y$ for the values $k = -6, 0, 6, 12$.

Functions of several variables

Definition

A **function of three variables** f is a rule that assigns to each ordered triple (x, y, z) in a domain $D \subset \mathbb{R}^3$ a unique real number denoted by $f(x, y, z)$.

Example: The temperature T at a point on the surface of the earth depends on the longitude x and latitude y of the point and on the time t , so we could write $T = f(x, y, t)$.

Functions of several variables

- It's very difficult to visualize a function f of three variables by its graph, since that would lie in a four-dimensional space.
- However, we do gain some insight into f by examining its **level surfaces**

Definition

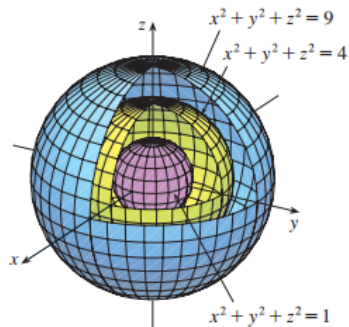
The **level surfaces** of a function f of three variables are the surfaces with equations $f(x, y, z) = k$, where k is a constant.

If the point (x, y, z) moves along a level surface, the value of $f(x, y, z)$ remains fixed.

Functions of several variables

Example: Find the level surfaces of the function

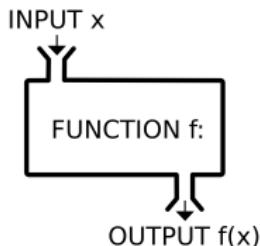
$$f(x, y, z) = x^2 + y^2 + z^2.$$



Functions of several variables

Definition

A **function of n variables** f is a rule that assigns to each incoming tuple of numbers, $(x_1, x_2, \dots, x_n)$, a uniquely defined outgoing number $z = f(x_1, x_2, \dots, x_n)$.



A subset of $\mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R} = \mathbb{R}^n$ called the n dimensional space.

Limits and Continuity

Definition

Let f be a function of two variables whose domain D includes points arbitrarily close to (a, b) . Then we say that **the limit of $f(x, y)$ as (x, y) approaches (a, b)** is L and we write

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$$

if for every number $\varepsilon > 0$ there is a corresponding number $\delta > 0$ such that

if $(x, y) \in D$ and $0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta$ then $|f(x, y) - L| < \varepsilon$.

Other notations for the limit in the above definition are

$$\begin{array}{l} \lim_{\substack{x \rightarrow a \\ y \rightarrow b}} f(x, y) = L \quad \text{and} \quad f(x, y) \rightarrow L \text{ as } (x, y) \rightarrow (a, b). \end{array}$$

Example: Prove that $\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2 + y^2} = 0$.

Properties of the limits

Let f and g be a function of two variables whose domain D includes points arbitrarily close to (a, b) . Assume that

- the limit of $f(x, y)$ as (x, y) approaches (a, b) is L_1 .
- the limit of $g(x, y)$ as (x, y) approaches (a, b) is L_2

Then, we have the following properties

- (i) $\lim_{(x,y) \rightarrow (a,b)} [f(x, y) \pm g(x, y)] = L_1 \pm L_2$.
- (ii) $\lim_{(x,y) \rightarrow (a,b)} [\alpha f(x, y)] = \alpha L_1$ with $\alpha \in \mathbb{R}$.
- (iii) $\lim_{(x,y) \rightarrow (a,b)} [f(x, y)g(x, y)] = L_1 L_2$.
- (iv) $\lim_{(x,y) \rightarrow (a,b)} \frac{f(x, y)}{g(x, y)} = \frac{L_1}{L_2}$ if $L_2 \neq 0$.

Limits and Continuity

The Squeeze Theorem

Let D be an domain containing the points arbitrarily close to (a, b) . Suppose that for every (x, y) in D not equal to (a, b) , we have

$$g(x, y) \leq f(x, y) \leq h(x, y)$$

and also suppose that

$$\lim_{(x,y) \rightarrow (a,b)} g(x, y) = \lim_{(x,y) \rightarrow (a,b)} h(x, y) = L.$$

Then,

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L.$$

Limits and Continuity

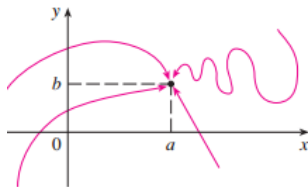
Example: Find $\lim_{(x,y) \rightarrow (0,0)} x \sin \left(\frac{y}{x} \right)$.

Limits and Continuity

- For functions of a single variable, when we let x approach a , there are only two possible directions of approach, from the left or from the right.

if $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$, then $\lim_{x \rightarrow a} f(x)$ does not exist.

- For functions of two variables the situation is not as simple because we can let (x, y) approach (a, b) from an infinite number of directions in any manner whatsoever as long as (x, y) stays within the domain of f .



$\Rightarrow$ if the limit exists, then $f(x, y)$ must approach the same limit no matter how (x, y) approaches (a, b) .

Limits and Continuity

Thus, if we can find two different paths of approach along which the function $f(x, y)$ has different limits, then it follows that $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ does not exist.

Theorem

If $f(x, y) \rightarrow L_1$ as $(x, y) \rightarrow (a, b)$ along a path C_1 and $f(x, y) \rightarrow L_2$ as $(x, y) \rightarrow (a, b)$ along a path C_2 , where $L_1 \neq L_2$, then $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ does not exist.

Example: Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ does not exist.

Limits and Continuity

Example: If $f(x, y) = \frac{xy}{x^2 + y^2}$, does $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exist?

Limits and Continuity

Definition

A function f of two variables is called **continuous at** (a, b) if

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b).$$

We say f is **continuous on** D if f is continuous at every point (a, b) in D .

Theorem

- Sums, differences, products, and quotients of continuous functions are continuous on their domains.
- If f and g are continuous functions on D then $g \circ f$ is continuous on D .

Remark: Any elementary function (the constant functions, the polynomial functions, the rational functions, the exponential functions, the logarithms functions, the trigonometric functions, the inverse trigonometric functions, the hyperbolic functions, the inverse hyperbolic functions,...) is continuous.

Example: Discuss the continuity of the function

$$f(x, y) = x^2y^3 - x^3y^2 + 3x + 2y$$

at $(1, 2)$.

Example: Discuss the continuity of the functions f and g are defined by

$$f(x, y) = \begin{cases} \frac{3x^2y}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0) \end{cases}$$
$$g(x, y) = \begin{cases} \frac{3x^2y}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0) \\ 2, & \text{if } (x, y) = (0, 0) \end{cases}$$

Definition

Let f be a function of two variables x and y

- The **partial derivative of f with respect to x at (a, b)** , denoted by $f_x(a, b)$, is obtained by keeping y fixed ($y = b$) and finding the ordinary derivative at a of the function $g(x) = f(x, b)$:

$$f_x(a, b) = \lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h}.$$

- The **partial derivative of f with respect to y at (a, b)** , denoted by $f_y(a, b)$, is obtained by keeping x fixed ($x = a$) and finding the ordinary derivative at b of the function $G(y) = f(a, y)$:

$$f_y(a, b) = \lim_{h \rightarrow 0} \frac{f(a, b + h) - f(a, b)}{h}.$$

Example: Let

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0) \end{cases}$$

Find $f_x(0, 0)$ and $f_y(0, 0)$.

Partial derivatives

There are many alternative notations for partial derivatives.

Notations for Partial Derivatives

If $z = f(x, y)$, we write

$$f_x(a, b) = \frac{\partial f}{\partial x}(a, b) = \frac{\partial z}{\partial x}(a, b) = f_1(a, b) = D_1 f(a, b) = D_x f(a, b).$$

$$f_y(a, b) = \frac{\partial f}{\partial y}(a, b) = \frac{\partial z}{\partial y}(a, b) = f_2(a, b) = D_2 f(a, b) = D_y f(a, b).$$

Rule for Finding Partial Derivatives of $z = f(x, y)$

- To find f_x , regard y as a constant and differentiate $f(x, y)$ with respect to x .
- To find f_y , regard x as a constant and differentiate $f(x, y)$ with respect to y .

Partial derivatives

Example: If $f(x, y) = x^3 + x^2y^3 - 2y^2$, find $f_x(2, 1)$ and $f_y(2, 1)$.

Higher Derivatives

If f is a function of two variables, then its partial derivatives f_x and f_y are also functions of two variables, so we can consider their partial derivatives $(f_x)_x$, $(f_x)_y$, $(f_y)_x$ and $(f_y)_y$, which are called the **second partial derivatives** of f . If $z = f(x, y)$, we use the following notation:

$$(f_x)_x = f_{xx} = f_{11} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 z}{\partial x^2}$$

$$(f_x)_y = f_{xy} = f_{12} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 z}{\partial y \partial x}$$

$$(f_y)_x = f_{yx} = f_{21} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 z}{\partial x \partial y}$$

$$(f_y)_y = f_{yy} = f_{22} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 z}{\partial y^2}.$$

Remark: The notation f_{xy} means that we first differentiate with respect to x and then with respect to y , whereas in computing f_{yx} the order is reversed.

Example: Find the second partial derivatives of

$$f(x, y) = x^3 + x^2y^3 - 2y^2.$$

Clairaut's Theorem

Suppose f is defined on a disk D that contains the point (a, b) . If the functions f_{xy} and f_{yx} are both **continuous** on D then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

Partial derivatives of order 3 or higher can also be defined. For instance

$$f_{xyy} = (f_{xy})_y = \frac{\partial}{\partial y} \left(\frac{\partial^2 f}{\partial y \partial x} \right) = \frac{\partial^3 f}{\partial y^2 \partial x}.$$

and using Clairaut's Theorem it can be shown that $f_{xyy} = f_{yxy} = f_{yyx}$ if these functions are continuous.

Higher Derivatives

Example: Calculate f_{xxy} if $f(x, y) = \sin(3x + y)$.

Tangent Planes

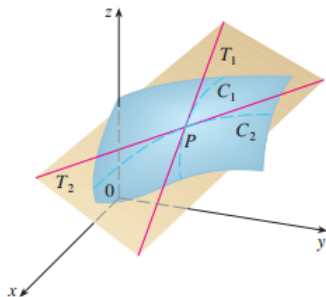


Figure: The tangent plane contains the tangent lines T_1 and T_2 .

Suppose a surface S has equation $z = f(x, y)$, where f has continuous first partial derivatives, and let

- $P(x_0, y_0, z_0)$ be a point on S
- C_1 and C_2 be the curves obtained by intersecting the vertical planes $y = y_0$ and $x = x_0$ with the surface S .

$\Rightarrow$ The point P lies on both C_1 and C_2 .

Tangent Planes

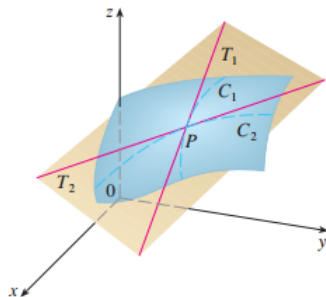


Figure: The tangent plane contains the tangent lines T_1 and T_2 .

Let T_1 and T_2 be the tangent lines to the curves C_1 and C_2 at the point P . Then, the **tangent plane** to the surface S at the point P is defined to be the plane that contains both tangent lines T_1 and T_2 .

Tangent Planes

Definition

Suppose f has continuous partial derivatives. An equation of the **tangent plane** to the surface $z = f(x, y)$ at the point $P(x_0, y_0, z_0)$ is

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

Example: Find the tangent plane to the elliptic paraboloid $z = 2x^2 + y^2$ at the point $(1, 1, 3)$.

Tangent Planes

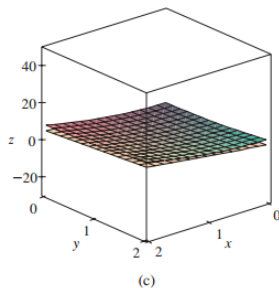
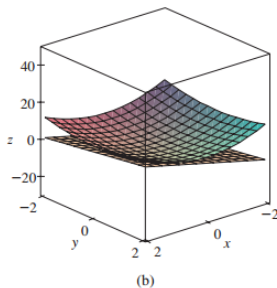
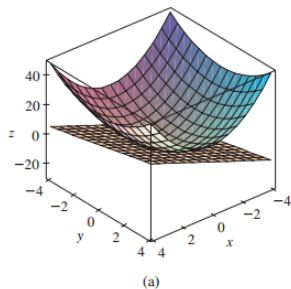


Figure: The elliptic paraboloid $z = 2x^2 + y^2$ appears to coincide with its tangent plane as we zoom in toward $(1, 1, 3)$.

Tangent Planes

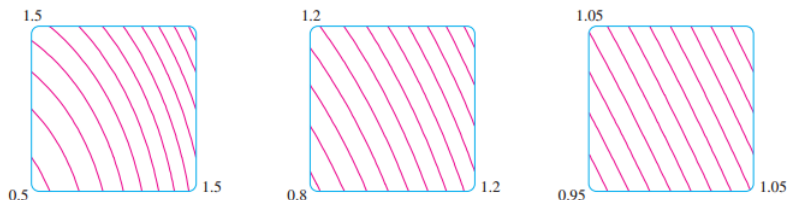


Figure: Zooming in toward $(1, 1)$ on a contour map of $f(x, y) = 2x^2 + y^2$.

In the above Figure, we corroborate this impression by zooming in toward the point $(1, 1)$ on a contour map of the function $f(x, y) = 2x^2 + y^2$. Notice that the more we zoom in, the more the level curves look like equally spaced parallel lines, which is characteristic of a plane.

Linear Approximations

In general, an equation of the tangent plane to the graph of a function f of two variables at the point $(a, b, f(a, b))$ is

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

The linear function whose graph is this tangent plane, namely

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

is called the **linearization** of f at (a, b) and the approximation

$$f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b) \quad (10)$$

is called the **linear approximation** or the **tangent plane approximation** of f at (a, b) .

Linear Approximations

Example: Let $f(x, y) = 2x^2 + y^2$. Find the linearization of f at $(1, 1)$.

Linear Approximations

Now consider a function of two variables, $z = f(x, y)$, and suppose x changes from a to $a + \Delta x$ and y changes from b to $b + \Delta y$. Then the corresponding **increment** of z is

$$\Delta z = f(a + \Delta x, b + \Delta y) - f(a, b).$$

$\Rightarrow$ The increment Δz represents the change in the value of f when (x, y) changes from (a, b) to $(a + \Delta x, b + \Delta y)$.

Linear Approximations

We define the differentiability of a function of two variables as follows

Definition

If $z = f(x, y)$ then f is **differentiable** at (a, b) if Δz can be expressed in the form

$$\Delta z = f_x(a, b)\Delta x + f_y(a, b)\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y$$

where ε_1 and $\varepsilon_2 \rightarrow 0$ as $(\Delta x, \Delta y) \rightarrow (0, 0)$.

Theorem (Sufficient condition for differentiability)

If the partial derivatives f_x and f_y exist near (a, b) and are continuous at (a, b) , then f is differentiable at (a, b) .

Linear Approximations

Example: Show that $f(x, y) = xe^{xy}$ is differentiable at $(1, 0)$ and find its linearization there. Then use it to approximate

Definition

For a differentiable function of two variables, $z = f(x, y)$, we define the **differentials** dx and dy to be independent variables; that is, they can be given any values. Then the **differential** dz , also called the **total differential**, is defined by

$$dz = f_x(x, y)dx + f_y(x, y)dy = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy. \quad (11)$$

If we take $dx = \Delta x = x - a$ and $dy = \Delta y = y - b$, then the differential of z is

$$dz = f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

So, in the notation of differentials, the linear approximation (10) can be written as

$$f(x, y) \approx f(a, b) + dz.$$

Example:

- a) If $z = f(x, y) = x^2 + 3xy - y^2$, find the differential dz .
- b) If x changes from 2 to 2.05 and y changes from 3 to 2.96, compare the values of Δz and dz .

The Chain Rule

The Chain Rule (Case 1)

Suppose that $z = f(x, y)$ is a differentiable function of x and y , where $x = g(t)$ and $y = h(t)$ are both differentiable functions of t . Then z is a differentiable function of t and

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

Since we often write $\frac{\partial z}{\partial x}$ in place of $\frac{\partial f}{\partial x}$, we can rewrite the Chain Rule in the form

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}.$$

The Chain Rule

Example: If $z = x^2y + 3xy^4$, where $x = \sin(2t)$ and $y = \cos(t)$, find dz/dt when $t = 0$.

The Chain Rule

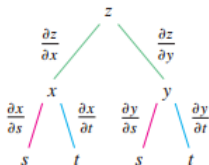
The Chain Rule (Case 2)

Suppose that $z = f(x, y)$ is a differentiable function of x and y , where $x = g(s, t)$ and $y = h(s, t)$ are both differentiable functions of s and t . Then

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

and

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}.$$



The Chain Rule

Example: If $z = e^x \sin(y)$, where $x = st^2$ and $y = s^2t$, find $\partial z / \partial s$ and $\partial z / \partial t$.

The Chain Rule

The Chain Rule (General Version)

Suppose that u is a differentiable function of the n variables $x_1, x_2, \dots, x_n$ and each x_j is a differentiable function of the m variables $t_1, t_2, \dots, t_m$. Then u is a function of $t_1, t_2, \dots, t_m$ and

$$\frac{\partial u}{\partial t_i} = \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_i} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial t_i} + \dots + \frac{\partial u}{\partial x_n} \frac{\partial x_n}{\partial t_i}$$

for each $i = 1, 2, \dots, m$.

The Chain Rule

Example: If $u = x^4y + y^2z^3$, where $x = rse^t$, $y = rs^2e^{-t}$, and $z = r^2s \sin(t)$, find the value of $\partial u / \partial s$ when $r = 2$, $s = 1$, $t = 0$.

Implicit Differentiation

- We suppose that an equation of the form $F(x, y) = 0$ defines y implicitly as a differentiable function of x , that is, $y = f(x)$, where $F(x, f(x)) = 0$ for all x in the domain of f .
- If F is differentiable, we can apply Case 1 of the Chain Rule to differentiate both sides of the equation $F(x, y) = 0$ with respect to x . So, we obtain

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} \frac{dy}{dx} = 0.$$

- If $\frac{\partial F}{\partial y} \neq 0$ then we solve for $\frac{dy}{dx}$ and obtain

$$\frac{dy}{dx} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}} = -\frac{F_x}{F_y}. \quad (12)$$

Implicit Differentiation

Example: Find $\frac{dy}{dx}$ if $x^3 + y^3 = 6xy$.

Implicit Differentiation

- Now, we suppose that z is given implicitly as a function $z = f(x, y)$ by an equation of the form $F(x, y, z) = 0$. This means that $F(x, y, f(x, y)) = 0$ for all (x, y) in the domain of f .
- If F and f are differentiable, then we can use the Chain Rule to differentiate the equation $F(x, y, z) = 0$ with respect to x . So, we obtain

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x} = 0.$$

- If $\frac{\partial F}{\partial z} \neq 0$ then we solve for $\frac{\partial z}{\partial x}$ and obtain

$$\frac{\partial z}{\partial x} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}} = -\frac{F_x}{F_z}. \quad (13)$$

- The formula for $\frac{\partial z}{\partial y}$ is obtained in a similar manner

$$\frac{\partial z}{\partial y} = -\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}} = -\frac{F_y}{F_z}. \quad (14)$$

Implicit Differentiation

Example: Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if $x^3 + y^3 + z^3 + 6xyz = 1$.

Directional derivative

Definition

The **directional derivative** of f at (x_0, y_0) in the direction of a **unit vector** $\mathbf{u} = \langle a, b \rangle$ (i.e. $a^2 + b^2 = 1$) is

$$D_{\mathbf{u}}f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}$$

if this limit exists.

Theorem

If f is a differentiable function of x and y , then f has a directional derivative in the direction of any unit vector $\mathbf{u} = \langle a, b \rangle$ and

$$D_{\mathbf{u}}f(x, y) = f_x(x, y)a + f_y(x, y)b. \quad (15)$$

Example: Find the directional derivative $D_{\mathbf{u}}f(x, y)$ if

$$f(x, y) = x^3 - 3xy + 4y^2$$

and $\mathbf{u}$ is the unit vector given by angle $\theta = \pi/6$. What is $D_{\mathbf{u}}f(1, 2)$?

The Gradient Vector

Definition

If f is a function of two variables x and y , then the **gradient** of f is the vector function ∇f defined by

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle .$$

With this notation for the gradient vector, we can rewrite equation (15) for the directional derivative of a differentiable function as

$$D_u f(x, y) = \nabla f(x, y) \cdot \mathbf{u} \tag{16}$$

The Gradient Vector

Example: Find the directional derivative of the function $f(x, y) = x^2y^3 - 4y$ at the point $(2, -1)$ in the direction of the vector $v = \langle 2, 5 \rangle$.

Maximizing the Directional Derivative

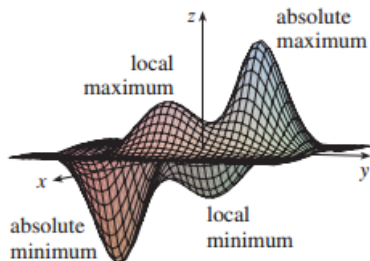
Theorem

Suppose f is a differentiable function of two variables. The **maximum value** of the **directional derivative** $D_{\mathbf{u}}f(x, y)$ is $|\nabla f(x, y)|$ and it occurs when $\mathbf{u}$ has the same direction as the gradient vector $\nabla f(x, y)$.

Example:

- a) If $f(x, y) = xe^y$, find the rate of change of f at the point $P(2, 0)$ in the direction from P to $Q\left(\frac{1}{2}, 2\right)$.
- b) In what direction does f have the maximum rate of change? What is this maximum rate of change?

Extremum of functions of several variables

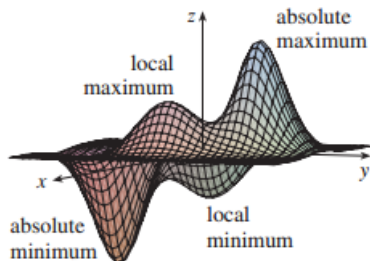


Definition

Let f is a function of two variables.

- If $f(x, y) \leq f(a, b)$ when (x, y) is near (a, b) , then f has a **local maximum** at (a, b) and $f(a, b)$ is a **local maximum value**.
- If $f(x, y) \leq f(a, b)$ hold **for all** points (x, y) in the domain of f , then f has a **absolute maximum** at (a, b) .

Extremum of functions of several variables



Definition

Let f is a function of two variables.

- If $f(x, y) \geq f(a, b)$ when (x, y) is near (a, b) , then f has a **local minimum** at (a, b) and $f(a, b)$ is a **local minimum value**.
- If $f(x, y) \geq f(a, b)$ hold **for all** points (x, y) in the domain of f , then f has a **absolute minimum** at (a, b) .

Extremum of functions of several variables

Theorem

If f has a local maximum or minimum at (a, b) and the first-order partial derivatives of f exist there, then $f_x(a, b) = 0$ and $f_y(a, b) = 0$.

Definition

A point (a, b) is called a **critical point** (or *stationary point*) of f if $f_x(a, b) = 0$ and $f_y(a, b) = 0$, or if one of these partial derivatives does not exist.

$\Rightarrow$ if f has a local maximum or minimum at (a, b) , then (a, b) is a critical point of f .

Is the reverse of the above reason correct?

Extremum of functions of several variables

Example: Find the extreme values of $f(x, y) = y^2 - x^2$.

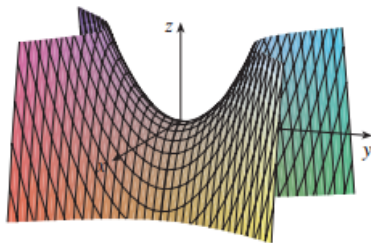


Figure: The graph of $f(x, y) = y^2 - x^2$.

Extremum of functions of several variables

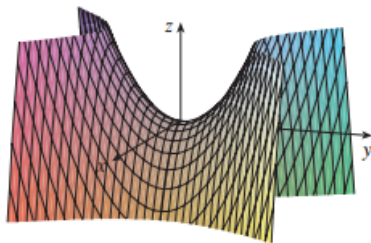


Figure: The graph of $f(x, y) = y^2 - x^2$.

You can see that $f(0, 0) = 0$ is a maximum in the direction of the x-axis but a minimum in the direction of the y-axis. Near the origin the graph has the shape of a saddle and so $(0, 0)$ is called a **saddle point** of f .

Extremum of functions of several variables

We need to be able to determine whether or not a function has an extreme value at a critical point by using the following test

Second Derivatives Test

Suppose the second partial derivatives of f are continuous on a disk with center (a, b) , and suppose that (a, b) is a critical point of f . Let

$$D = D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2.$$

- a) If $D > 0$ and $f_{xx}(a, b) > 0$, then $f(a, b)$ is a local minimum.
- b) If $D > 0$ and $f_{xx}(a, b) < 0$, then $f(a, b)$ is a local maximum.
- c) If $D < 0$, then $f(a, b)$ is not a local maximum or minimum. In this case, the point (a, b) is called a saddle point of f .

Extremum of functions of several variables

Example: Find the local maximum and minimum values and saddle points of $f(x, y) = x^4 + y^4 - 4xy + 1$.

Extremum of functions of several variables

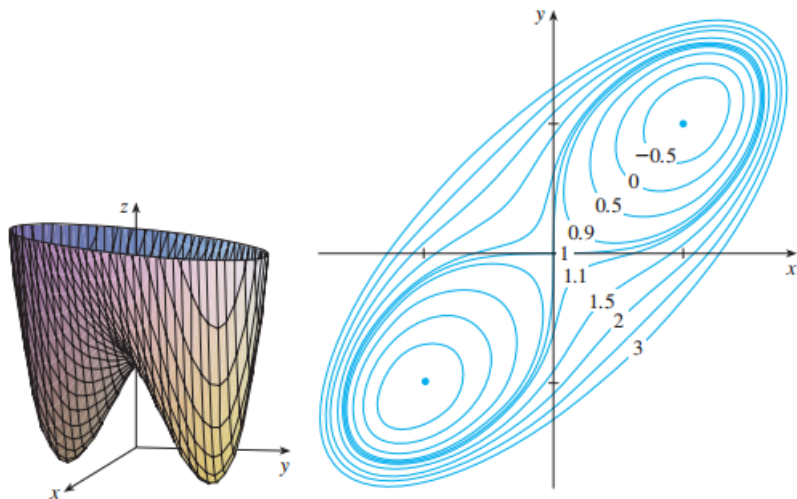


Figure: The graph and a contour map of $f(x, y) = x^4 + y^4 - 4xy + 1$.

Absolute Maximum and Minimum Values

Definition

- A **boundary point** of D is a point (a, b) such that every disk with center (a, b) contains points in D and also points not in D .
- A **closed set** in $\mathbb{R}^2$ is one that contains all its boundary points.
- A **bounded set** in $\mathbb{R}^2$ is one that is contained within some disk. In other words, it is finite in extent.



(a) Closed sets



(b) Sets that are not closed

Absolute Maximum and Minimum Values

Extreme Value Theorem for Functions of Two Variables

If f is continuous on a closed, bounded set D in $\mathbb{R}^2$, then f attains an absolute maximum value $f(x_1, y_1)$ and an absolute minimum value $f(x_2, y_2)$ at some points (x_1, y_1) and (x_2, y_2) in D .

The Closed Interval Method

To find the absolute maximum and minimum values of a continuous function f on a closed, bounded set D :

1. Find the values of f at the critical points of f in D .
2. Find the extreme values of f on the boundary of D .
3. The largest of the values from steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

Absolute Maximum and Minimum Values

Example: Find the absolute maximum and minimum values of the function

$$f(x, y) = x^2 - 2xy + 2y$$

on the rectangle $D = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 3, 0 \leq y \leq 2\}$

Lagrange multipliers

Method of Lagrange Multipliers

To find the maximum and minimum values of $f(x, y)$ subject to the constraint $g(x, y) = k$ (assuming that these extreme values exist and $\nabla g \neq 0$ on the surface $g(x, y) = k$):

- a) Find all values of x, y and λ such that

$$\begin{cases} \nabla f(x, y) = \lambda \nabla g(x, y) \\ g(x, y) = k. \end{cases}$$

- b) Evaluate f at all the points (x, y) that result from step a). The largest of these values is the maximum value of f . The smallest is the minimum value of f .

Note that: The number λ is called a **Lagrange multiplier**.

Lagrange multipliers

Example: Find the extreme values of the function $f(x, y) = x^2 + 2y^2$ on the circle $x^2 + y^2 = 1$.

THANK YOU FOR YOUR
ATTENTION !